2 The following measurements were taken of a uniform metal disc.

mass $m = (265.2 \pm 0.5)$ g

diameter $D = (10.3 \pm 0.1)$ cm

thickness $t = (1.36 \pm 0.02)$ cm

electrical resistance $R = (20 \pm 1)\ \Omega$

The density ρ of the disc is determined using the above measurements. Which measurement contributes the most to the uncertainty of the value of ρ?

A m **B** D **C** t **D** R

Ans: **B**

$$\rho = \frac{m}{\left(\frac{\pi D^2}{4} \times t\right)} \rightarrow \pm\frac{\Delta\rho}{\rho} = \pm\left(\frac{\Delta m}{m} + 2\frac{\Delta D}{D} + \frac{\Delta t}{t}\right) = \pm(0.19\% + 2 \times 0.97\% + 1.5\%)$$

3 A spring obeying Hooke's law has an unstretched length of 45 mm and a spring constant of